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# RoboHarness: A Simple Harness Outperforms VLA and World Action Models

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https://github.com/BinceQu/RoboHarness

https://bincequ.github.io/RoboHarness

The rise of large language models (LLMs) has opened a path toward generalist embodied AI, raising the question of how LLMs can act in the physical world. One straightforward approach is training the model to generate actions: collect robot-action data and video demonstration, refine LLM architecture with an action head, and train a vision-language-action (VLA) model at scale. However, we argue that coding agents have the potential to realize generalist embodied manipulation. All they need is a simple yet effective robot harness. We present **RoboHarness**: a robotic harness that gives LLM agents a visual-geometric control panel so that they can directly understand and invoke embodied tasks. With RoboHarness, Qwen3.8-Flash-Next (Qwen Team, 2026) outperforms state-of-the-art VLA models on BEHAVIOR Challenge 2025 tasks (Li et al., 2024). The implementation is open source.

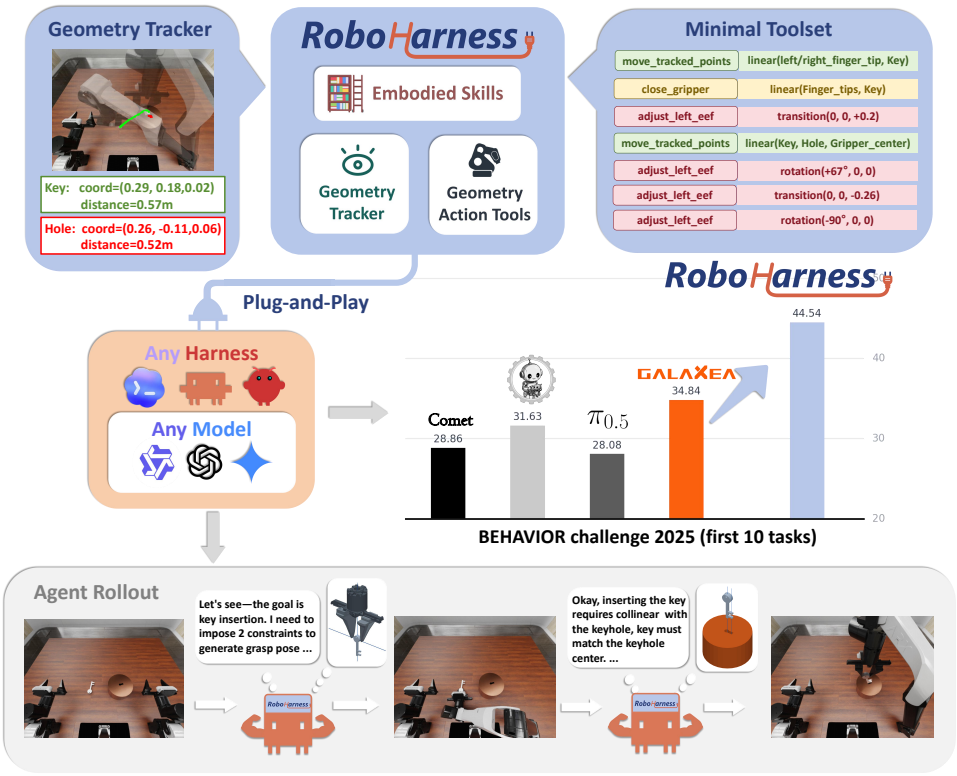



Figure 1: RoboHarness plugs into coding-agent harnesses such as Claude Code, tracks keypoints as geometric constraints, and verifies execution in a closed loop. On the first 10 BEHAVIOR Challenge 2025 tasks, Qwen3.8-Flash-Next attains a higher mean $q$-score than the state-of-the-art VLA models shown.

# 1 Introduction

The emergence of large language models (LLMs) opened a path towards generalist robot manipulation. The community has explored many ways to make use of LLM for planning, visual understanding, physical reasoning, and spatial reasoning to achieve generalizable robot manipulation. Yet an LLM's world knowledge lives in a textual space. A bridge between the LLM and the physical world is therefore required:

One straightforward approach is training the model to generate actions. VLA models try to make the model output end-effector actions directly. Actions may be encoded into the vocabulary for autoregressive generation, as in RT-2, OpenVLA, and FAST (Brohan et al., 2023; Kim et al., 2025b; Pertsch et al., 2025), or grouped into flow-matching chunks, which are later produced by an action head appended to the LLM (Black et al., 2025b; Bjorck et al., 2025). This representation can express arbitrary robot actions, but it requires substantial robot-action data and additional post-training. Such data are difficult to collect and scale to the diversity of real-world objects, scenes, and tasks. This difficulty has motivated world-action models (WAMs), which seek to improve generalization by learning action-conditioned models of how the world changes (Zhao et al., 2026; Wang et al., 2026; Li et al., 2026).

We take a different view. An LLM already has strong capabilities for task decomposition, planning, spatial reasoning, and tool use. The main challenge is therefore not to train the LLM to generate a larger set of actions, but to provide a simple and sufficiently complete interface that connects these capabilities to the physical world.

Existing work provides evidence for this view: Packaging embodied subtasks, such as picking, placing, and navigation, as callable functions allows an LLM to compose them into longer procedures (Liang et al., 2023). ASPIRE shows that an LLM agent can refine such programs and accumulate the repairs in a skill library (Lu et al., 2026). However, the available functions are not complete: their coverage is determined by predefined tasks, objects, and policies. Every new object or task may require another function, leaving the LLM confined to a predefined action space. This suggests that the main limitation lies in the action interface rather than in the LLM's ability.

In this paper, we propose **RoboHarness**. Without training models, RoboHarness combines visual clicks, keypoint tracking, and geometric constraints to provide an embodied control panel for an LLM agent. Concretely, our contributions are:

**An expressive, LLM-friendly robot harness** that provides an embodied control panel for an LLM agent.

**A new paradigm for embodied AI.** We show that a generalist LLM, equipped with the right tool harness can outperform VLA and World Action Models

With RoboHarness, the open-source model Qwen3.8-Flash-Next can directly solve long-horizon household tasks in BEHAVIOR-1K (Li et al., 2024) without any finetune, outperforming state-of-the-art models. It also substantially outperforms VLA and WAM baselines on fully out-of-distribution generalization tests.

# 2 Related Work

Prior work connects pretrained language and vision-language models to manipulation through learned action generation, executable programs, or spatial representations. These approaches differ in where they place the boundary between model reasoning and robot control.

## 2.1 Vision-Language-Action Models

**Autoregressive action tokens.** RT-2 and OpenVLA adapt pretrained VLMs to predict discretized robot actions through next-token prediction (Brohan et al., 2023; Kim et al., 2025b). FAST improves this approach by compressing action chunks with a frequency-domain tokenizer, making autoregressive learning more effective for high-frequency control (Pertsch et al., 2025). Galaxea G0.5 further integrates reasoning and action tokens into one autoregressive stream, combining task decomposition, visual grounding, and motor commands within a shared model (Liu et al., 2026).

**Continuous action experts.** Another family couples a pretrained VLM with a specialized module for generating continuous action sequences. Representative examples include the flow-matching action expert in $\pi_0$ and the diffusion transformer in GR00T N1 (Black et al., 2025b; Bjorck et al., 2025). More recent models, such as $\pi_{0.7}$, condition action generation on richer task context, including subtask instructions, subgoal images, and episode metadata, to improve steerability (Physical Intelligence, 2026). These approaches learn a broad motor action space, but require robot-action supervision and additional training. Preserving pretrained knowledge is also a training-design concern: knowledge insulation mitigates interference by blocking action-expert gradients from entering the VLM backbone (Driess et al., 2025). RoboHarness connects an unchanged LLM to robot control through a geometric tool interface, avoiding this additional model-training stage.

### 2.2 LLMs as Robot Planners and Programmers

**Skill and module orchestration.** SayCan grounds language-level skill selection in learned affordance values, while Inner Monologue incorporates environmental feedback into replanning (Ahn et al., 2023; Huang et al., 2023b). HarnessVLA combines a frozen VLA for contact-rich manipulation with analytic primitives, using execution memory to guide retries and recovery (Zhang et al., 2026).

**Program synthesis and skill discovery.** Code as Policies extends composition to generated control programs, including feedback loops and waypoint-level commands (Liang et al., 2023). CaP-X systematically varies the abstraction level of robot APIs and finds that performance degrades as human-designed abstractions are removed; execution feedback and skill discovery help agents work with lower-level interfaces (Fu et al., 2026). ASPIRE compounds program repairs into a growing skill library (Lu et al., 2026).

### 2.3 Spatial Representations for Manipulation

**Visual points and motion proposals.** RoboPoint learns to predict image-space affordance points from synthetic supervision (Yuan et al., 2025). PIVOT iteratively selects among visual action proposals with a frozen VLM, and MOKA selects marked keypoints and waypoints (Nasiriany et al., 2024; Fang et al., 2024). HAMSTER predicts a two-dimensional end-effector path and passes it to a separate low-level policy (Li et al., 2025). VoxPoser grounds language in composable three-dimensional value maps and uses model-based planning to generate six-degree-of-freedom end-effector trajectories (Huang et al., 2023a). Constraint-based approaches express the required geometry more explicitly. CoPa grounds object parts in the observed scene and solves geometric constraints over those parts (Huang et al., 2024). OmniManip represents interaction points and directions in an object's canonical space and tracks the resulting six-dimensional poses (Pan et al., 2025). ReKep synthesizes relational constraints over three-dimensional keypoints as Python functions and optimizes subgoals and paths in a closed perception–action loop (Huang et al., 2025). These works establish that grounded geometric relations can support orientation-aware, multi-stage manipulation without task-specific VLM training.

## 3 Method

Lessons from coding agents show that: a minimal set of simple, orthogonal primitive tools outperforms a collection of predefined features and heavy scaffolding. Inspired by this principle, we argue that an LLM's potential on embodied tasks is best unlocked by giving it the minimal primitive action tools rather than layers of engineered features.

We start from studying spatial relations between a gripper and objects it interacts with, therefore constructing the most elementary primitive action tools. We further prove their completeness in Section 4: (i) **any contact pose** between a gripper and an object can be expressed by geometric constraints, and (ii) once the gripper holds an object, **any spatial relationship** between the in-hand object and a target object can be expressed by geometric constraints.

With the primitive action tools in place, the remaining challenge is spatial understanding: although VLMs cannot reliably reason about 3D spatial relations, their grounding training enables them to localize specific points on objects with reasonable accuracy. We therefore use spatial grounding as the bridge between vision-language models and robots. The LLM marks points of interest on a 2D

image and continuously receives their positions via optical-flow tracking and depth back-projection; with this information, it composes accurate primitive actions to perform complex manipulation.

Following this design, we build a simple but effective Robot Harness, which we will release as open source to the community.

### 3.1 Geometry State for Spatial Grounding

To make primitive action tools more usable to the LLM, we design a persistent geometry-state tracking mechanism that continuously returns the coordinates of objects of interest to the LLM, enhancing its accurate spacial perception capabilities.

#### 3.1.1 Robot coordination definition

We first define a unified camera-frame coordination for diverse robot configurations, making it straightforward for the LLM to read spatial relations:

| Axis | Positive direction |
|---|---|
| $x$ | forward from camera view |
| $y$ | left from camera view |
| $z$ | upward from camera view |

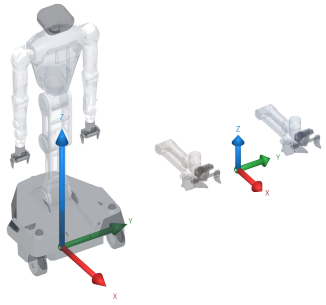


Figure 2: Visualization of the unified coordinate frame on two embodiments: BEHAVIOR R1 Pro (Li et al., 2024) (left) and RoboDojo ARX X5 (Chen et al., 2026) (right).

#### 3.1.2 Track keypoints

With a unified coordinate frame in place, we design the `track_object` tool for LLM to retrieve coordinates of points in which it is interested using no simulation priors the procedure and proceeds as follows:

1. The LLM selects one or more points of interest that are visible in the head camera.
2. Harness continuously tracks each marked point across successive head-camera frames.
3. The latest 2D location is projected onto the depth view to recover the metric distance.
4. The pixel coordinate and depth are then back-projected into the robot camera frame.

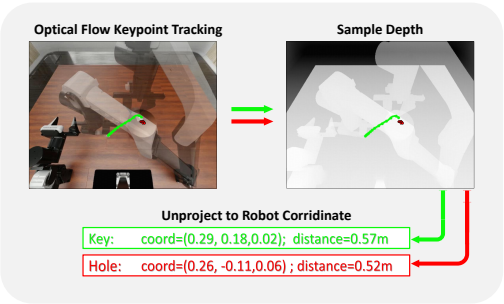



Figure 3: Pipeline of the `track_object` tool: optical-flow keypoint tracking followed by depth back-projection into the robot coordinate frame.

### 3.2 Action Toolset

Our goal is to design a minimal action toolset.

Therefore, we start from object-interaction relations and build the toolset bottom-up.

A manipulation task can be viewed as a two-stage process: first the gripper establishes a spatial relationship with an object, and then an end-effector action drives this object to satisfy a geometric relationship with another object. For instance, in an insert-key task, the gripper first grasps the key, then moves it until it satisfies the insertion constraint with the lock. Consequently, our action toolset only needs to capture these two stages.

We implement the first stage with keypoint-and-constraint tools: the LLM selects points on the image and specifies the geometric constraints they must satisfy; the harness automatically solves and executes the corresponding actions. To support model fine-tuning, we additionally provide adjustment tools that directly command the robot's low-level degrees of freedom.

In the next section, we will prove that, for any object, any grasp pose can be expressed by two keypoint constraints when the object has not yet been grasped; and when it has already been grasped, assuming the gripper forms a rigid connection with the object, the desired spatial relation between this held object and any other object in the scene can be expressed by two sets of keypoint constraints. Therefore, our action toolset is both minimal and complete.

#### 3.2.1 Keypoint-constraint tools for manipulation

The model supplies a set of input points together with their spatial relations, and the harness automatically solves and executes the corresponding action. The model must distinguish between on-hand and off-hand points: on-hand points lie on the gripper or on a held object and their positions can be modified, whereas off-hand points are fixed in the scene. The harness drives the on-hand points so that they satisfy the spatial constraints specified by the model through the following pipeline:

1. Sample action candidates and select those satisfying the constraints.
2. Discard candidates whose inverse kinematics are infeasible or unreachable.
3. Plan a motion trajectory from the current pose to the target pose.
4. Execute the trajectory.

In principle, the model can express any constraint through coordinate relations among the input points. However, expressions such as collinearity and coplanarity are both verbose to write and frequently required, so we encapsulate the most common patterns as quick requests; their specific modes are listed in Table 1.

Table 1: Modes for keypoint-constraints tool

| **Mode** | **Input points** | **Description** |
|---|---|---|
| coordinate | *arbitrary* | move points to satisfy the given coordinate constraints |

| **Quick-requests** | **On-hand** | **Off-hand** | **Description** |
|---|---|---|---|
| touch | 1 | 1 | an on-hand point coincides with an off-hand point |
| linear | 2 | 1 or 2 | two on-hand points collinear with off-hand points |
| line-vertical-plane | 2 | 3 | an on-hand line is orthogonal to an off-hand plane |
| plane-parallel | 3 | 3 | on-hand plane parallel to off-hand plane |
| line-vertical-ground | 2 | 0 | on-hand line vertical to the ground |
| face to | 3 | 0 | face the target surface |
| move to reach | 0 | 1 | move chassis to make target reachable |
| grasp | 0 | 1 | move chassis to make target reachable |

#### 3.2.2 Adjustment tools

To refine robot action or to improve robustness when the constraints fail (for example, when the object slips out of the gripper), we also expose direct low-level robot control interfaces. These interfaces share the same coordinate frame defined in Section 3.1.1, so that the LLM can interpret and use them consistently.

Table 2: Adjustment tools. Displacements are incremental; chassis `spin` is a yaw increment.

| Name | Parameters | Meaning |
|---|---|---|
| adjust left eef pose | $(\Delta x, \Delta y, \Delta z, r, p, y)$ | increment the left end-effector pose or rotation |
| adjust right eef pose | $(\Delta x, \Delta y, \Delta z, r, p, y)$ | increment the right end-effector pose or rotation |
| adjust chassis | (fwd, side, spin) | move the chassis forward, sideways or spin |
| adjust height | $\Delta h$ | raise or lower the body |
| adjust pitch | $\Delta\theta$ | change the body pitch |

## 4 Proof: Completeness of the Keypoint-Constraint Representation

The three points on the robot gripper, namely the left fingertip, right fingertip, and gripper center, are intrinsic points that remain tracked throughout execution (Figure 4.1). The following results show that constraining their geometric relationships can represent arbitrary motion trajectories and interaction poses. The elementary correspondence between a contact pose and a keypoint triple is given in Section 4.1.

### 4.1 Any Grasp Pose Can Be Expressed by Keypoint Constraints

**Definition:** The gripper carries three predefined points $\{F_L, F_R, G\}$: the left fingertip, the right fingertip, and the slider center. A contact pose is $(f_l, f_r, \theta)$ with $f_l, f_r \in \partial A$. We prove that $(f_l, f_r, \theta)$ is in analytic bijection with three surface points $(a_1, a_2, a_3)$ under the constraints overlap$[a_1, F_L]$, overlap$[a_2, F_R]$, and linear$[c, a_3, G]$, where $c = \frac{a_1+a_2}{2}$.

**Proposition 1.** *The gripper carries predefined points $\{F_L, F_R, G\}$ and a contact pose is $(f_l, f_r, \theta)$ with $f_l, f_r \in \partial A$. We prove an analytic bijection to three surface points $(a_1, a_2, a_3)$ under* overlap$[a_1, F_L]$, overlap$[a_2, F_R]$, *and* linear$[c, a_3, G]$, *where $c = \frac{a_1+a_2}{2}$.*

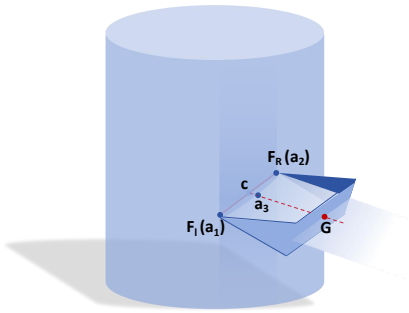



Figure 4: Three predefined gripper points $\{F_L, F_R, G\}$ and a contact pose $(f_l, f_r, \theta)$.

*Proof.*

**(i)** $(f_l, f_r, \theta) \mapsto (a_1, a_2, a_3)$. From the two overlap constraints, $a_1 = f_l$ and $a_2 = f_r$. From linear, $a_3$ satisfies

$$a_3 = c + k\big(p(G) - c\big), \qquad a_3 \in \partial A,\ k > 0. \tag{1}$$

Hence $(a_1, a_2, a_3)$ is uniquely determined.

**(ii)** $(a_1, a_2, a_3) \mapsto (f_l, f_r, \theta)$. From overlap, $f_l = a_1$ and $f_r = a_2$. Let $u := (a_2 - a_1)/\|a_2 - a_1\|$, fix a right-handed orthonormal basis $(\mathbf{v}_1, \mathbf{v}_2)$ of $u^\perp$, and set $h := \|p(G) - c\|$. Then

$$p(G, \theta) = c + h\big(\mathbf{v}_1 \cos\theta + \mathbf{v}_2 \sin\theta\big), \qquad h\big(\mathbf{v}_1 \cos\theta + \mathbf{v}_2 \sin\theta\big) \times (a_3 - c) = \mathbf{0}. \tag{2}$$

Since $(a_3 - c) \perp (a_1 - a_2)$, the cross-product condition yields a unique $\theta$:

$$\tan\theta = \frac{(a_3 - c) \cdot \mathbf{v}_2}{(a_3 - c) \cdot \mathbf{v}_1}. \tag{3}$$

The two maps are mutual inverses, so $(a_1, a_2, a_3)$ and $(f_l, f_r, \theta)$ are in analytic bijection. □

□

### 4.2 Any Relationship Between In-Hand and Off-hand Object Can Be Expressed by Keypoint Constraints

**Proposition 2.** *For any prescribed relative coordinate transform $T_{\text{true}} \in SE(3)$ between two rigid objects, there exist two non-parallel spatial line-coincidence constraints such that any unknown $T \in SE(3)$ satisfying the resulting system is uniquely determined, and the solution is $T_{\text{true}}$.*

*Proof.*
*Existence.* Place the objects at $T_{\text{true}}$. In the global frame pick two non-parallel lines $\ell_1, \ell_2$, with body-$A$ parameters $L_{A1}, L_{A2}$ and body-$B$ parameters $L_{B1}, L_{B2}$. Rigid kinematics gives

$$T_{\text{true}}(L_{B1}) = L_{A1}, \qquad T_{\text{true}}(L_{B2}) = L_{A2}. \quad (4)$$

Hence such a pair of characteristic line correspondences exists.

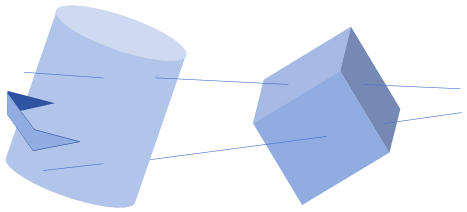


Figure 5: Two non-parallel line coincidences fix a unique $T \in SE(3)$.

*Reduction.* Let $T \in SE(3)$ satisfy the same two coincidences. Align local frames so that the $Z$ axis of $A$ coincides with $L_{A1}$ and the $Z$ axis of $B$ with $L_{B1}$. Then $T(L_{B1}) = L_{A1}$ forces $T$ to preserve the $Z$ axis, so $T$ retains two degrees of freedom: a rotation $\theta$ about $Z$ and a translation $d$ along $Z$. For $\mathbf{p} = (x, y, z)^\top$ on $B$, the image $\mathbf{p}' = (x', y', z')^\top$ is $x' = x\cos\theta - y\sin\theta,\ y' = x\sin\theta + y\cos\theta,\ z' = z + d$,

$$x' = x\cos\theta - y\sin\theta, \quad y' = x\sin\theta + y\cos\theta, \quad z' = z + d. \quad (5)$$

*Unique $\theta$.* The second constraint is $T(L_{B2}) = L_{A2}$. Non-parallelism implies $L_{B2} \nparallel Z$. Let $\mathbf{v}_{xy} = (v_x, v_y)^\top \neq \mathbf{0}$ be the $XY$ projection of the direction of $L_{B2}$, and $\mathbf{u}_{xy} = (u_x, u_y)^\top$ that of $L_{A2}$. Rotation about $Z$ preserves the projected length, $\|\mathbf{v}_{xy}\| = \|\mathbf{u}_{xy}\|$, and

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} u_x \\ u_y \end{pmatrix}. \quad (6)$$

The determinant is nonzero, so this system has a unique real solution $\theta \in [0, 2\pi)$.

*Unique $d$.* Pick a known point $\mathbf{p}_0 = (x_0, y_0, z_0)^\top$ on $L_{B2}$. Substituting $x_0, y_0$ and the solved $\theta$ into (5) yields constants $x_0', y_0'$. The image of $\mathbf{p}_0$ lies on $L_{A2}$, and $L_{A2} \nparallel Z$, so $(x_0', y_0')$ determines a unique height $Z_{\text{target}}$ on $L_{A2}$. The third identity in (5) gives $z_0 + d = Z_{\text{target}}$, hence $d = Z_{\text{target}} - z_0$.

Thus $T$ is uniquely solved and equals $T_{\text{true}}$. □

## 5 Experiments

### 5.1 RoboHarness can outperform VLA and WAM across tasks

**BEHAVIOR 2025: Long-horizon household tasks.** We compare against state-of-the-art VLAs such as Galaxea G0.5 and $\pi_{0.5}$, evaluating both success rate and subgoal completion rate (Liu et al., 2026; Black et al., 2025a). We additionally include the first- and second-place winners of the BEHAVIOR Challenge 2025 (RLC and Comet, respectively) (Larchenko et al., 2025; Bai et al., 2025), whose $q$-scores are reported on the official leaderboard. The RoboHarness column is Qwen3.8-Flash-Next. Each entry is the mean $q$-score over five instances per task. Due to the limited evaluation budget, we only run five instances per task with seed `20260911`, using slots $0, 3, 5, 7, 9$ (instance IDs 301, 304, 306, 308, 310). The prompt is specific to each task but shared across instances, and is written by a human. The maximum tick count is set separately for each task, and the ten limits are listed in Appendix A. Per-instance scores and the rollout archive are in Appendix A.

Table 3: BEHAVIOR Challenge 2025 task results. The RoboHarness column is Qwen3.8-Flash-Next, and each entry is the five-instance mean in Appendix A.

| ID | Task | RLC | Comet | $\pi_0$.5 4ep | G0.5 1ep | RoboHarness |
|---|---|---|---|---|---|---|
| 0 | turning on radio | 0.6000 | 1.0000 | 0.1500 | 0.3000 | 0.4000 |
| 1 | picking up trash | 0.6667 | 0.7667 | 0.5500 | 0.8167 | 0.8667 |
| 2 | putting away Halloween decorations | 0.2000 | 0.5000 | 0.3714 | 0.4857 | 0.6857 |
| 3 | cleaning up plates and food | 0.1857 | 0.0000 | 0.3857 | 0.1786 | 0.2571 |
| 4 | can meat | 0.0000 | 0.0000 | 0.0333 | 0.0111 | 0.0000 |
| 5 | setting mousetraps | 0.3333 | 0.0000 | 0.1083 | 0.5083 | 0.8333 |
| 6 | hiding Easter eggs | 0.1444 | 0.2444 | 0.0778 | 0.0500 | 0.4222 |
| 7 | picking up toys | 0.3000 | 0.0000 | 0.1833 | 0.3167 | 0.3667 |
| 8 | rearranging kitchen furniture | 0.3000 | 0.3750 | 0.3875 | 0.3500 | 0.4000 |
| 9 | putting up Christmas decorations inside | 0.4333 | 0.0000 | 0.5611 | 0.4667 | 0.2222 |
| Percentage | | 31.635% | 28.861% | 28.084% | 34.838% | 44.539% |

### 5.2 Does the RoboHarness Generalize across Objects?

**Objects.** The catalog is the 100 rigid objects in Figure 6. Assets are taken from the BEHAVIOR object library (Li et al., 2024), have extent at most 0.21 m, and have mass at most 1.2 kg. `saucepan` and one recorded outlier are kept outside those two cuts.

**Tasks.** The task is to pick up each of the 100 objects, and to place each object in the embodiment and simulation environment on which the corresponding model was trained. For ASPIRE and RoboHarness, the object is placed on the floor and the robot starts from a stooping pose. For models trained on LIBERO (Liu et al., 2023), the LIBERO tabletop scene is cleared and the object is placed at the center of the table. Success conditions:

| | |
|---|---|
| Max ticks | 2000 |
| Success | $\max z - z_0 \geq 5\,\text{cm}$ |



Figure 6: The 100 objects used to evaluate generalization across objects. The catalog is drawn from the BEHAVIOR object library.

**Analysis and results.**

**Generalization is not positively correlated with the amount of robot data.** In Figure 7, Faster-WAM reaches 61/100 from a few hours of robot demonstrations, while several models trained on thousands of hours score lower (Zhao et al., 2026). The three world action models initialized from Wan2.2, Faster-WAM, OpenWAM-$\alpha$, and LingBot-VA, sit above most vision-language-action models despite very different robot-data budgets (Wan Team, 2025; Wang et al., 2026; Li et al., 2026).

**Multimodal capabilities matter more than model parameter count.** The RoboHarness points use no robot training set and still score 68–86. These points sit above the learned policies. Among those policies, the smaller world action models outscore larger vision-language-action models such as OpenVLA-OFT (Kim et al., 2025a). Per-object outcomes are in Appendix D.

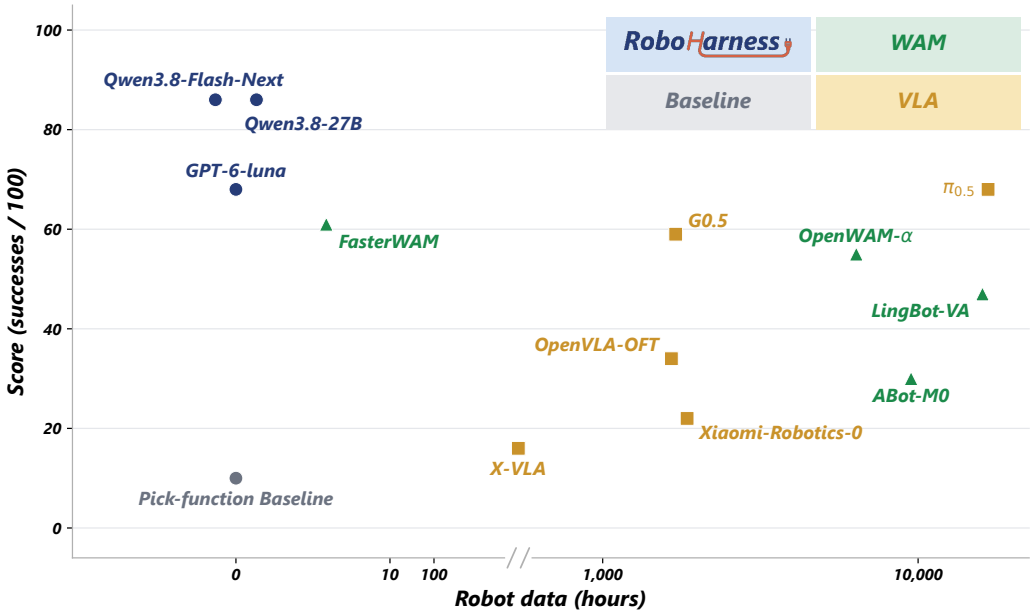



Figure 7: Successes out of 100 objects versus estimated hours of robot data. Colors: VLA (blue), world action model (green), RoboHarness (red), and the ASPIRE-pick baseline (gray). The hour conversion is in Appendix C. Qwen3.8-Flash-Next and Qwen3.8-27B both have 0 robot-data hours. Their markers are drawn 0.14 axis units left and right of 0, respectively, so the two points remain visible.

## 6 Limitation & Discussion

This work explores a new technical path for generalist Embodied AI, constructing a simple but effective harness design that lets an LLM directly command a robot, and experimentally shows that this approach substantially outperforms both VLA and WAM baselines.

LLMs can come up with brilliant plans and solutions for embodied tasks, but often fail to ground in the robot's physical embodiment, leading to failures such as knocking over objects, falling, or getting stuck. To mitigate this on BEHAVIOR long-horizon mobile manipulation tasks, we hand-craft a task-specific prompt for every task that specifies the execution steps and the behavioral boundaries the model must respect.

Future work aims to close this gap further by training the model or evolving the harness to become more physically grounded.

### AI use statement

Generative AI tools were used in three ways. (i) They assisted writing. (ii) They assisted running the experiments: launching jobs, monitoring long rollouts, and flagging failed runs so that each experiment kept running stably, while every experimental setting and every reported number was checked by the authors. (iii) They assisted coding: writing and debugging RoboHarness human interface.

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## A BEHAVIOR Five-Instance Scores

The RoboHarness column of Table 3 is a five-instance sample of each of the first ten BEHAVIOR Challenge 2025 tasks, and every rollout in that column uses Qwen3.8-Flash-Next. The robot is the R1 Pro embodiment of the challenge. The sample is fixed by seed `20260911`. Every task loads the same evaluation slots $0, 3, 5, 7, 9$, which are activity instances $301, 304, 306, 308, 310$. Slots $1, 2, 4, 6, 8$ (instances $302, 303, 305, 307, 309$) are outside this sample. Each instance is one official rollout. The prompt is specific to each task but shared across instances, and is written by a human. It includes a procedure for accomplishing the task. The official BEHAVIOR Challenge 2025 evaluator scores the rollout, and a cell in Table 5 is `q_score.final` from that evaluation. The task mean is the unweighted average of those five scores, rounded to four decimals, and is the number reported in Table 3.

An episode ends when the task goal is already satisfied or when the simulator reaches the maximum tick count of that task. The maximum is the official 2025 limit for the task, set to twice the mean length of its human demonstrations, and it is counted in simulator control steps. Table 4 lists the limit used for each task.

Table 4: Official maximum tick count for each BEHAVIOR Challenge 2025 task in this sample. The episode stops at this count unless the goal is already satisfied.

| ID | Task | Max ticks |
|---|---|---|
| 0 | turning on radio | 4299 |
| 1 | picking up trash | 10535 |
| 2 | putting away Halloween decorations | 27664 |
| 3 | cleaning up plates and food | 27392 |
| 4 | can meat | 23694 |
| 5 | setting mousetraps | 20343 |
| 6 | hiding Easter eggs | 15239 |
| 7 | picking up toys | 37781 |
| 8 | rearranging kitchen furniture | 17886 |
| 9 | putting up Christmas decorations inside | 27437 |

Table 5: Per-instance $q$-score for the five-instance RoboHarness sample. Slots $0, 3, 5, 7, 9$ are instances $301, 304, 306, 308, 310$.

| ID | Task | 301 slot 0 | 304 slot 3 | 306 slot 5 | 308 slot 7 | 310 slot 9 | Mean |
|---|---|---|---|---|---|---|---|
| 0 | turning on radio | 0.0000 | 0.0000 | 0.0000 | 1.0000 | 1.0000 | 0.4000 |
| 1 | picking up trash | 1.0000 | 0.3333 | 1.0000 | 1.0000 | 1.0000 | 0.8667 |
| 2 | putting away Halloween decorations | 0.5714 | 0.7143 | 0.4286 | 0.7143 | 1.0000 | 0.6857 |
| 3 | cleaning up plates and food | 0.2857 | 0.2857 | 0.2857 | 0.1429 | 0.2857 | 0.2571 |
| 4 | can meat | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| 5 | setting mousetraps | 1.0000 | 1.0000 | 0.5000 | 1.0000 | 0.6667 | 0.8333 |
| 6 | hiding Easter eggs | 1.0000 | 0.0000 | 0.7778 | 0.3333 | 0.0000 | 0.4222 |
| 7 | picking up toys | 0.0000 | 0.6667 | 0.5000 | 0.6667 | 0.0000 | 0.3667 |
| 8 | rearranging kitchen furniture | 0.2500 | 0.5000 | 0.2500 | 0.5000 | 0.5000 | 0.4000 |
| 9 | putting up Christmas decorations inside | 0.3333 | 0.2222 | 0.3333 | 0.1111 | 0.1111 | 0.2222 |
| Task 0–9 mean | | | | | | | 0.44539 |

## B Full Embodied Skill Catalog

We distill the procedures used across individual tasks into a set of reusable embodied skills, listed in Table 6.

## C Evaluation Setting and Robot-Data Hours

**Test.** Each learned policy is evaluated on a cleared LIBERO table. The object is placed at the center, the prompt is `pick up the {name}`, and the trial runs for 2000 steps. The trial succeeds when the

Table 6: Embodied skills shipped with the `embodied_claude_code` plugin, excluding the trivial single-object skills `close-box`, `cut-object`, and `stand-trash-can-upright`. Each row is condensed to one sentence describing what the skill does.

| Skill | Description |
|---|---|
| `behavior-v2-baseline` | contract of scene perception, task decomposition, primitive selection, and execution monitoring. |
| `navigate-to-target` | Drives the chassis toward a marked target. |
| `open-doors-and-drawers` | Open hinged doors or drawers. |
| `pick-up-object` | Plans and executes a grasp of a visible object. |
| `place-object-in-container` | Aligns a held object over an open container and releases it only after the object has settled inside. |
| `traverse-narrow-passages` | Drives chassis through tight doorway . |

target root rises by at least 5 cm. Qwen3.8-Flash and gpt-6-luna use the RoboHarness on the R1 Pro, with the object on the floor and the same success rule. An hour counts robot recordings and human egocentric manipulation video once. Web-video generation and VQA are not counted.

**Qwen3.8-27B.**

- The run calls the served model `Qwen3.8-27B-FP8` with the skill prompt. This model has no released robot checkpoint on Hugging Face. The trial uses the shared 2000-step protocol and the 5 cm lift.
- The run has no robot manipulation training set, so the robot-data hours are 0.

**Qwen3.8-Flash.**

- The run calls the served model `Qwen3.8-Flash-Next-FP8` with the skill prompt. This model has no released robot checkpoint on Hugging Face.
- The run has no robot manipulation training set, so the robot-data hours are 0.

**GPT-6-luna.**

- The run calls the served model `gpt-6-luna` with the skill prompt. This model has no released robot checkpoint on Hugging Face.
- The run has no robot manipulation training set, so the robot-data hours are 0.

$\pi_{0.5}$.

- The run uses the released `pi05_libero` checkpoint from the openpi release. The policy replans every 5 actions.
- The paper states about 400 hours of mobile-manipulator data, and states that 97.6% of the examples in the first training phase come from other robots or the web (Black et al., 2025a). Converting the whole first-phase mixture by that share gives 16,666.7 hours. The paper does not separate the web portion from the other robot data.

**Faster-WAM.**

- The run uses `hustvl/FasterWAM`, the released LIBERO weights at step 21700. The policy replans every 10 actions.
- The paper initializes the model from Wan2.2-5B, marks embodied pretraining as absent, and trains on four LIBERO suites of 500 demonstrations for 10 epochs (Zhao et al., 2026; Wan Team, 2025). The training release `yuanty/LIBERO-fastwam` states 1,712 episodes and 277,713 frames at 20 frames per second, which converts to 3.86 hours (Yuan, 2026). The Wan corpus is web video and is not counted (Wan Team et al., 2025).

**Galaxea G0.5.**

- The run uses the `g05-libero` checkpoint released in `OpenGalaxea/G05`. The policy predicts 10 action steps at a time.
- The G0.5 paper states 14 embodiments and does not state a training duration (Liu et al., 2026). The estimate therefore uses stage 1 of the previous G0 paper, which states about 1,000 hours of Open X-Embodiment, 500 hours of Galaxea Open-World, and 200 hours collected in-house (Jiang et al., 2025; Open X-Embodiment Collaboration, 2024). The no-op-filtered LIBERO demonstrations used for this checkpoint convert to 3.80 hours at 20 Hz. Adding these durations gives 1,703.8 hours.

**OpenWAM-$\alpha$.**

- The run uses `OpenWAM/OpenWAM-Alpha-Sim-LIBERO` at step 10690. Denoising uses 10 steps, and the inference horizon is 10 actions.
- The paper lists 1,442 hours of egocentric human video, 1,794 hours of AgiBot, 686 hours of RoboCOIN, 1,007 hours of DROID, and 1,440 hours of InternData-A1, and states that this mixture is seen for one epoch (Wang et al., 2026; Wu et al., 2025b; Tian et al., 2025). Adding these durations gives 6,369 hours. The Wan2.2 initialization is web video and is not counted (Wan Team et al., 2025; Wan Team, 2025).

**LingBot-VA.**

- The run uses `robbyant/lingbot-va-posttrain-libero-long`. The trial uses the shared setting of 2000 steps and a 5 cm lift.
- The paper states approximately 16K hours of robot manipulation data, and states that pretraining consumes 1.4T tokens (Li et al., 2026). The plotted value is the stated 16,000 hours. The token count records how much data the run consumes, and it is not added again as a second copy of the corpus.

**OpenVLA-OFT.**

- The run uses `moojink/openvla-7b-oft-finetuned-libero-10`. The unnormalization key is `libero_10_no_noops`, and the client executes the full 8-step action chunk.
- The OpenVLA paper states 970k demonstrations from 27 Open X-Embodiment datasets, trained for 27 epochs, with the DROID portion removed for the last third of training (Kim et al., 2025b; Open X-Embodiment Collaboration, 2024). The released frame counts of those 27 datasets convert to 1,651.4 hours. The OFT paper fine-tunes this checkpoint on LIBERO and does not add another pretraining corpus (Kim et al., 2025a).

**ABot-M0.**

- The run uses `acvlab/ABot-M0-LIBERO` at step 40000. The action chunk is 10 steps, and the unnormalization key is `franka`.
- The paper states that UniACT contains more than six million trajectories and 9,500 hours (Yang et al., 2026). The plotted value is the stated 9,500 hours.

**Xiaomi-Robotics-0.**

- The run uses `XiaomiRobotics/Xiaomi-Robotics-0-LIBERO`. The task id is `libero_all`, and the policy replans every 10 actions.
- The paper states about $200 \times 10^6$ robot timesteps on a unified 30 Hz timeline (Cai et al., 2026). These timesteps convert to 1,851.9 hours. The 338 hours and 400 hours of in-house tasks stated in the paper are already inside this total.

**X-VLA.**

- The run uses `2toINF/X-VLA-Libero`. The domain id is 3, and denoising uses 10 steps.

- The pretraining table lists episode counts: AgiBot has 141K episodes, DROID is listed as two rows of 45K episodes, and RoboMIND is listed as 19K, 2K, 25K, and 11K episodes (Zheng et al., 2026). AgiBot states 1,001,552 trajectories and 2,976.4 hours, which converts to 419.0 hours for the listed 141K episodes (Bu et al., 2025). DROID states 76k successful trajectories and 350 hours, and the two 45K rows are one subset of that release, which converts to 207.2 hours (Khazatsky et al., 2024). RoboMIND states 107k trajectories and 305.5 hours for the full corpus, which converts to 162.7 hours for the listed 57K episodes (Wu et al., 2025a). Adding these three durations gives 788.9 hours. The sampling weights in the X-VLA table are mixture weights, and they are not used as durations.

**ASPIRE pick function.**

- The baseline has no learned checkpoint. Each object is scored by one call of the ASPIRE pick function, and the same 5 cm lift is a success (Lu et al., 2026).
- The baseline has no robot manipulation training set, so the robot-data hours are 0.

## D Per-Object Outcomes

Table 7 lists the runs in Figure 7. A check is a success and a cross is a failure under the 5 cm rule of Section 5.2.

Table 7: Per-object outcome on the 100 objects. A check is a success under the 5 cm rule and a cross is a failure. Columns are Qwen3.8-27B, Qwen3.8-Flash, gpt-6-luna, $\pi_{0.5}$, Faster-WAM, G0.5, OpenWAM-$\alpha$, LingBot-VA, OpenVLA-OFT, ABot-M0, Xiaomi-Robotics-0, X-VLA, and the ASPIRE pick function.

| # | Object | 27B | Flash | luna | $\pi_{0.5}$ | F-WAM | G0.5 | OWAM | Ling | OVLA | ABot | Xiao | XVLA | Pick |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | alarm_clock | ✓ | × | × | × | × | × | × | × | × | × | × | × | × |
| 2 | apple | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | × | × | × |
| 3 | avocado | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | × | × | ✓ | × | × |
| 4 | banana | ✓ | ✓ | ✓ | × | ✓ | × | × | × | × | × | × | ✓ | × |
| 5 | bar_soap | ✓ | × | × | ✓ | ✓ | ✓ | ✓ | × | × | × | × | × | × |
| 6 | baseball | × | ✓ | × | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × |
| 7 | battery | ✓ | ✓ | ✓ | × | × | × | ✓ | ✓ | ✓ | × | × | × | × |
| 8 | bell_pepper | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × |
| 9 | blackboard_eraser | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | × | × |
| 10 | calculator | × | × | × | × | × | ✓ | × | × | × | × | × | × | × |
| 11 | can_of_soda | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × | × | × | ✓ |
| 12 | canned_food | ✓ | ✓ | × | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | ✓ | ✓ | × |
| 13 | carrot | ✓ | ✓ | × | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × |
| 14 | cell_phone | ✓ | ✓ | × | ✓ | × | × | × | × | × | × | × | × | ✓ |
| 15 | cellulose_tape_dispenser | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × | × | × | × |
| 16 | coffee_cup | ✓ | × | × | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | × |
| 17 | cucumber | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | ✓ | ✓ | × | × |
| 18 | deodorant_stick | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | × | × | × | × | ✓ |
| 19 | dental_floss | × | ✓ | ✓ | × | ✓ | × | ✓ | × | × | × | × | × | × |
| 20 | digital_camera | ✓ | ✓ | × | × | × | × | × | × | × | × | × | × | × |
| 21 | duct_tape | ✓ | ✓ | ✓ | × | × | × | × | × | × | × | × | × | ✓ |
| 22 | easter_egg | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × | × | × |
| 23 | egg | × | × | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × |
| 24 | eggplant | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × | ✓ | × | × |
| 25 | garlic | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | ✓ | × |
| 26 | gaming_controller | ✓ | ✓ | ✓ | × | × | × | × | × | × | × | × | × | × |
| 27 | hockey_puck | ✓ | ✓ | × | ✓ | ✓ | × | ✓ | ✓ | × | × | × | × | × |
| 28 | inhaler | ✓ | ✓ | × | × | × | ✓ | ✓ | ✓ | ✓ | × | × | × | × |
| 29 | lemon | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | × | × | × |
| 30 | lime | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × |
| 31 | lipstick | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | ✓ | × | × | × | × |
| 32 | mouse | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × | ✓ | × | × | × |
| 33 | mug | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × | × | ✓ |
| 34 | orange | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | ✓ | × | × | × |
| 35 | paper_cup | ✓ | ✓ | × | ✓ | × | ✓ | ✓ | × | × | × | × | ✓ | × |
| 36 | peach | ✓ | ✓ | ✓ | ✓ | × | × | × | ✓ | ✓ | ✓ | × | × | × |
| 37 | pear | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | ✓ | × | × | × |
| 38 | pepper_grinder | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × | ✓ | × | × |
| 39 | pill_bottle | ✓ | ✓ | × | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × | × | × |
| 40 | pillar_candle | ✓ | ✓ | × | × | × | × | × | × | × | × | × | × | × |
| 41 | plum | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × | × | × |
| 42 | pool_ball | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × |
| 43 | portable_game_console | × | ✓ | × | × | × | ✓ | × | ✓ | × | × | × | × | × |
| 44 | potato | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | × | ✓ | × |
| 45 | salt_shaker | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × | × |
| 46 | soda_cup | × | ✓ | ✓ | ✓ | × | × | × | × | × | ✓ | × | ✓ | ✓ |
| 47 | stapler | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × | × | ✓ | × | × |
| 48 | tennis_ball | ✓ | × | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × |

| # | Object | 27B | Flash | luna | $\pi_{0.5}$ | F-WAM | G0.5 | OWAM | Ling | OVLA | ABot | Xiao | XVLA | Pick |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 49 | toy_dice | ✓ | ✓ | ✓ | × | ✓ | × | ✓ | ✓ | ✓ | × | × | × | × |
| 50 | wallet | ✓ | ✓ | × | ✓ | × | ✓ | × | × | × | × | × | × | × |
| 51 | artichoke | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × | × | × |
| 52 | baby_bottle | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × | × | × |
| 53 | beaker | ✓ | ✓ | ✓ | ✓ | × | × | × | ✓ | × | × | × | ✓ | × |
| 54 | beefsteak_tomato | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × | ✓ | × | ✓ | × |
| 55 | beer_glass | ✓ | ✓ | ✓ | ✓ | × | × | ✓ | × | × | × | ✓ | × | × |
| 56 | beet | ✓ | ✓ | × | × | × | × | × | × | × | × | × | × | × |
| 57 | bottle_of_hot_sauce | ✓ | ✓ | × | ✓ | × | ✓ | × | ✓ | × | × | ✓ | × | × |
| 58 | bowl | ✓ | ✓ | × | ✓ | × | × | × | × | × | × | × | × | ✓ |
| 59 | brussels_sprouts | ✓ | ✓ | × | ✓ | ✓ | ✓ | ✓ | × | ✓ | × | × | × | × |
| 60 | can_of_beans | ✓ | ✓ | × | ✓ | × | × | × | × | × | × | × | ✓ | × |
| 61 | can_of_sardines | ✓ | ✓ | × | ✓ | × | ✓ | ✓ | × | × | × | × | ✓ | × |
| 62 | canteen | ✓ | ✓ | ✓ | × | × | × | × | × | × | × | × | × | × |
| 63 | carton_of_milk | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | ✓ | × | × | × | × | × |
| 64 | fried_chicken | ✓ | ✓ | ✓ | × | × | × | × | × | × | × | × | × | × |
| 65 | cinnamon_roll | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × | × |
| 66 | marker | × | ✓ | × | ✓ | ✓ | ✓ | ✓ | × | ✓ | × | × | × | × |
| 67 | conch | ✓ | ✓ | ✓ | × | × | × | × | × | × | × | × | × | × |
| 68 | corkscrew | ✓ | ✓ | ✓ | × | ✓ | ✓ | × | ✓ | × | × | ✓ | × | × |
| 69 | gravy_boat | ✓ | ✓ | ✓ | × | × | × | × | × | × | × | × | × | × |
| 70 | eyeglasses | ✓ | × | × | × | × | × | × | × | × | × | × | × | ✓ |
| 71 | french_press | ✓ | × | ✓ | × | × | × | × | × | × | × | × | × | × |
| 72 | gift_box | × | ✓ | ✓ | × | × | × | × | × | × | × | × | × | ✓ |
| 73 | glue_stick | × | ✓ | ✓ | × | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × |
| 74 | grapefruit | ✓ | ✓ | ✓ | × | × | × | × | × | × | × | × | × | × |
| 75 | hard_drive | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | × | × | × | × | × |
| 76 | hotdog | ✓ | ✓ | ✓ | ✓ | × | ✓ | × | ✓ | × | × | × | × | × |
| 77 | spray_can | ✓ | × | ✓ | ✓ | × | ✓ | × | × | × | × | × | × | × |
| 78 | jade_roller | × | × | ✓ | × | ✓ | ✓ | ✓ | ✓ | × | × | × | × | × |
| 79 | jar_of_jam | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | ✓ | × | ✓ | ✓ | ✓ | × |
| 80 | jigger | ✓ | ✓ | ✓ | ✓ | ✓ | × | × | ✓ | × | × | × | × | × |
| 81 | light_bulb | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | ✓ | × | × | × | × | × |
| 82 | lighter | ✓ | ✓ | ✓ | × | × | ✓ | × | × | × | × | × | × | × |
| 83 | mango | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | × | × | × |
| 84 | match_box | ✓ | ✓ | × | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | × |
| 85 | mousetrap | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | × | × | × | × |
| 86 | pencil_case | ✓ | ✓ | ✓ | ✓ | × | ✓ | × | × | × | × | × | × | × |
| 87 | plier | × | ✓ | × | × | ✓ | ✓ | ✓ | × | × | × | × | × | × |
| 88 | pomegranate | ✓ | × | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | × | ✓ | × |
| 89 | razor | × | × | × | × | ✓ | ✓ | × | ✓ | × | × | × | × | × |
| 90 | saucepan | ✓ | × | × | × | × | × | × | × | × | × | × | × | × |
| 91 | screwdriver | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | × | ✓ | × | × |
| 92 | scrub_brush | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × |
| 93 | snow_globe | ✓ | ✓ | × | ✓ | × | × | × | ✓ | × | ✓ | × | × | × |
| 94 | teacup | × | ✓ | × | ✓ | × | × | ✓ | × | × | ✓ | ✓ | ✓ | × |
| 95 | tube_of_toothpaste | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | ✓ |
| 96 | vegetable_peeler | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × |
| 97 | water_glass | × | × | ✓ | ✓ | ✓ | × | × | ✓ | × | × | × | × | × |
| 98 | toy_figure | ✓ | ✓ | ✓ | × | × | × | × | × | × | × | × | × | × |
| 99 | spirit_lamp | ✓ | ✓ | × | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × | ✓ | × |
| 100 | zucchini | ✓ | ✓ | × | ✓ | ✓ | ✓ | ✓ | × | × | ✓ | × | × | × |